

MKTG100 Numeracy

Solutions To Week 8a's Exercises

AD

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Week 8a

1.2. Fixed cost and variable cost

Problem

The cost of making a one unit of a product is £40. What is the cost to make 20 units?

Answer:

£40 times 20 = £800

Exercise

Look at the table below.

Your Business (December)	
Item	Cost (£)
Variable costs per job	£50.00
Office rent	£1,500.00
Wages	£5,000.00
Advertising	£2,000.00
Other fixed costs	£800.00
Expected number of jobs per month	50

Calculate total cost.

Answer:

Fixed cost	£9,300.00	(Office rent + Wages + Advertising+ Other fixed costs)					
Variable cost	£2,500.00	(£50 * 50)					
Total cost	£11,800.00						

Figure 1: Total cost

Break-even analysis

Problem

You sell a product with a price of £20 per unit. The variable cost of the product is £10 per unit. The fixed costs associated with the product is £20,000. How many products should you sell to achieve break-even?

Answer:

Fixed cost = £20000

Selling price = £20

Variable cost per product = £10

$$\text{Contribution} = \text{Selling Price} - \text{Variable Cost} \quad (1)$$

$$= £20 - £10 \quad (2)$$

$$= £10 \quad (3)$$

Therefore,

$$BE \text{ quantity} = \frac{\text{Fixed cost}}{\text{Contribution}} \quad (4)$$

$$= \frac{£20000}{£10} \quad (5)$$

$$= 2000 \quad (6)$$

Thus, you should sell 2000 units.

Exercises

1. A company's variable costs per product are £5. The daily fixed costs are £100. If the product is sold at £10, calculate the break-even quantity.

Answer

$$\text{Contribution} = \text{Selling Price} - \text{Variable Cost} \quad (7)$$

$$= £10 - £5 \quad (8)$$

$$= £5 \quad (9)$$

Therefore,

$$\text{BE quantity} = \frac{\text{Fixed cost}}{\text{Contribution}} \quad (10)$$

$$= \frac{£100}{£5} \quad (11)$$

$$= 20 \quad (12)$$

2. If you increase the selling price by 5%, calculate the new break-even quantity.

Answer

$$\text{New selling price} = 105\% \times £10 = 10.5$$

$$\text{Contribution} = \text{Selling Price} - \text{Variable Cost} \quad (13)$$

$$= £10.5 - £5 \quad (14)$$

$$= £5.5 \quad (15)$$

Therefore,

$$\text{BE quantity} = \frac{\text{Fixed cost}}{\text{Contribution}} \quad (16)$$

$$= \frac{£100}{£5.5} \quad (17)$$

$$= 18.18 \quad (18)$$

The BE quantity is 18.18. You can not sell partial units. You may round up to the next whole number (if you round down, you will incur a profit loss). Thus, you must sell at least 19 units.